SET-A

End Term Exam, 2013-14 (Even Sem)

Soft Skills

Code- AHE-282

Course- B.Tech/MCA **Year- II** **Branch-All Branches**

Instructions :

- There are two sections in the question paper. Section I contains 40 questions, section II contains 70 questions
- Attempt all questions.
- The maximum marks for the test are 40.
- The total time allocated is 120 minutes. 000000000000000
- You have to write your answers **(1/2/3/4/5/A/B/C/D/E)** in the Answer sheet.
- There will be no negative marking.
- **Name of Student:-**

  Branch:-

  Section-

  University Roll No:
- **Signature of Invigilator-**

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1		23		45		67		89	
2		24		46		68		90	
3		25		47		69		91	
4		26		48		70		92	
5		27		49		71		93	
6		28		50		72		94	
7		29		51		73		95	
8		30		52		74		96	
9		31		53		75		97	
10		32		54		76		98	
11		33		55		77		99	
12		34		56		78		100	
13		35		57		79		101	
14		36		58		80		102	
15		37		59		81		103	
16		38		60		82		104	
17		39		61		83		105	
18		40		62		84		106	
19		41		63		85		107	
20		42		64		86		108	
21		43		65		87		109	
22		44		66		88		110	

Total Marks (I+II)-

Signature of Evaluator-

SECTION-I (40 questions, 80 marks)

1. If ABRIDGE is coded as BDUMIML, the code for CONCISE would be:
   (a) DQPGNZL (b) DQQGNYL (c) DQQHMYL (d) DQPGOYL (e) DQQGNYM
2. If according to a code FEUDALISM is EVFGZNTJM, then the code for CONTAINER would be:
   (a) UOPDBSFOJ (b) UOPDBSGOJ (c) UOPDZSFOK (d) UOPDZSFOJ (e) UOPDZSFPJ
3. If ENTRANCE is coded as FPVTBEF then the code for PANORAMA is:
   (a) RCPPTCOC (b) QBOPSBOB (c) QCQPTCOC (d) RBPPTBOB (e) None of these
4. If BRAIN is written as 12345 and DIVIDE is written as 647468 then BEREAVED would be written as:
   (a) 18273876 (b) 18283786 (c) 18283876 (d) 18283768(e) 18273678

Qs. 25-26. According to a code:
Jim went to Agra is 'lo po mo ro'
John never saw Agra is 'ro do co bo'
John belongs to Hyderabad is 'mo fo hodo'
Jim never liked Hyderabad is 'co qo fo lo'

5. What is the code for 'belongs'?
   (1) mo (2) fo (3) do (4) qo (5) ho
6. How would you write 'John liked Agra', in this code?
   (1) do qo lo (2) po ro mo (3) do qo ro (4) mo lo po (5) None of these
7. In a certain code PEAK is written as '3512' and DINE is written as '6895'. How is KIND written in that code?
   (1) 2396 (2) 2986 (3) 2896 (4) 2596 (5) None of these

8. In a certain code language 'JAT NO FIT ' means 'Ram and Shyam' , 'FAT HIT NO ' means 'Ram is intelligent ' and 'HIT JAT GUM ' means 'Shyam Is here ". Which of the following means 'here' in that code language?
   1) GUM 2) JAT 3) FAT 4) HIT

9. Ashish and Bhavesh, starting from the same place, make a journey of 56 km; Ashish starts 3 hours 20 minutes after Bhavesh, but travels 5 km an hour faster. If they arrive at the same time, what is Ashish's speed ?
   1] 13 kmph 2] 7 kmph 3] 12 kmph 4] 8 kmph

10. In racing a distance 'd' at uniform speed, A can beat B by 27 m. B can beat C by 87.5 m, and A can beat C by 104 m. What is the distance d ?
   1] 200 m 2] 250 m 3] 150 m 4] none of these

11. The distance between Charu's and Mani's places is 120 km. Charu travelled the whole distance from her place to Mani's at 30 km/hr but returned at 40 km/hr. Her average speed for the whole journey is (approximately):1)36km/hr 2) 37km/hr 3)34km/hr 4)33km/hr 5)35km/hr

12. A bus left Delhi for Ambala at 50 km/hr and turned over the same route at 40 km/hr. Thus it took 1 hour more on the return trip. The distance between Delhi and Ambala is:
   1)200 kms 2)180 kms 3)400 kms 4)None of these

13. A train running at 72 kms per hour crosses a coconut tree standing by the side of the track in 7 seconds. The length of the train is:
   1)104 metres 2)140 metres 3)504 metres 4)540 metres

14. A man travels three-fifths of distance AB at a speed of 3a, and the remaining at a speed of 2b. If he goes from B to A and back at a speed of 5c in the same time, then:
   [1] 1/a + 1/b = 1/c [2] a + b = c [3] 1/a + 1/b = 2/c [4] None of these

15. A person moves along the each edge of an equilateral triangle with speeds 5m/s,7m/s & 8m/s.What is the average speed of the enire journey ?
   (1)5.5 (2) 6.7 (3) 6.4 (4) 7.3

16. A is behind B by 60m. A & B are running at speed of 17m/s & 13m/s constantly. After how much time the distance between A & B will be 24m?
   (1)21 (2) 8 (3) 17 (4) None of these.

Venn –diagram
There are 80 players in a school who play one or more than one games. These games are Golf , volleyball & Cricket . If 40 play golf , 30 play cricket & 30 play volleyball. 15 play both Golf & cricket ,& 14 of them play both cricket & volleyball. Number of players who play Golf & volleyball is same as the number of players playing all the 3 games.

17. What is the maximum no. of players possible who play all the 3 games?
   (1)16 (2) 14 (3) 25 (3) 15

For Questions 28 to 30

There are 40 students who have passed in Maths , 50 in Physics & 60 in Chemistry . 20 students have passed in both Physics & Chemistry . 15 students have passed in both Chemistry & Maths . 18 students have passed in Maths & Physics.

18. What is the maximum no. of students who have passed in only Physics or only Chemistry ?
    1. 70 2. 60 3. 80 4. 90

19. What is the minimum no. of students who have passed in only Physics & Chemistry ?
    1. 7 2. 15 3. 80 4. 5

20. What is the minimum no. of students who have failed in only Chemistry ?
    1. 7 2. 6 3. 12 4. 3

21. A certain amount earns simple interest of Rs. 1750 after 7 years. Had the interest been 2% more, how much more interest would it have earned?
    a)Rs.35 b)Rs.245 b)Rs.350 d)Cannot be determined e)None of these

22. A person borrows Rs. 5000 for 2 years at 4% p.a. simple interest. He immediately lends it to another person at $6\frac{1}{4}$ p.a for 2 years. Find his gain in the transaction per year.
    a)Rs.112.50 b)Rs.125 c)Rs.150 d)Rs.167.50

23. The difference between simple interest and compound on Rs. 1200 for one year at 10% per annum reckoned half-yearly is:
    a)Rs.2.50 b) Rs.3 c) Rs. 3.75 d) Rs.4 e) None of these

24. The difference between compound interest and simple interest on an amount of Rs. 15,000 for 2 years is Rs. 96. What is the rate of interest per annum?
    a)8 b) 10 c) 12 d) cannot be determined e) None of these

25. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. The simple interest on the same sum for double the time at half the rate percent per annum is:
    a. 400 b) 500 c) 600 d) 800

26. A man took loan from a bank at the rate of 12% p.a. simple interest. After 3 years he had to pay Rs. 5400 interest only for the period. The principal amount borrowed by him was:
    a)Rs.2000 b)Rs.10000 c)Rs.15000 d)Rs.20000

27. A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. The rate of interest per annum is:
    a)5% b)8% c)12% d)15%

28. What will be the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and that for 9 years?
    a)1:3 b)1:4 c)2:3 d)Data Inadequate e)None of these

29. If A $ B means A is the brother of B; B * C means B is the son of C; C @ D means C is the wife of D and A # D means A is the son of D, how C is related to A?
    a)maternal grandmother b)maternal aunt c)aunt d)mother

30. Pointing to a girl Sandeep said, "She is the daughter of the only sister of my father." How is sandeep related to the girl?
    a)uncle b)father c)grandfather d)cousin

31. Pointing to a boy in the photograph Reena said, "He is the only son of the only child of my grandfather." How Reena is related to that boy?
    a)mother b)sister c)aunt d)cannot be determined

32. A * B means A is the sister of B. 2) A $ B means B is the mother of A. 3) A + B means A is the brother of B.4) A = B means B is the father of A. Which of the following means M is the maternal uncle of N?
    a)M= P + Q * N b) N + P = Q * M c) N * P $ Q * M d)none of these

Directions for Question (13 to 20) : A Cube is painted with color Red , Green & Blue on each pair of adjacent faces . Then this Cube is divided into 729 identical cubelets .

33. What are the minimum number of cuts required to divide the Cube into 729 cubelets?
    a)24 b) 27 c) 30 d) 21

34. How many cubelets are painted with exactly 2 colors ?
    a)37 b) 27 c) 84 d) 96 e) None of these

35. How many Cubelets are painted with 3 colors ?
    a)3 b) 2 c) 8 d) 6 e) None of these

36. How many Cubelets are painted with 2 colors & on 3 faces ?
    a)3 b) 2 c) 8 d) 6 e) None of these

37. How many Cubelets are painted with 1color & on 2 faces ?
    a)21 b) 63 c) 84 d) 96 e) None of these

38. How many Cubelets are painted with 1 Color ?

a)294 b) 315 c) 318 d) 296 e) None of these

39. How many Cubelets are painted with None of the colors ?
a)343 b) 243 c) 216 d) 512 e) None of these

40. How many Cubelets are painted with 1colors & on 1 face only ?
a)294 b) 315 c) 318 d) 296 e) None of these

SECTION-II (70 questions, 70 marks)

41. Which of the following constitutes interpersonal skills?

A) Self Awareness B) Assertiveness

C) Self Confidence D) Team Building

42. When you are assertive, you are actually-

A) Being violent B) Being forceful

C) Aiming for a win-win situation D) Making others feel good

43. Self awareness in a person implies-

A) The ability to learn from your mistakes.

B) Knowing most of your strengths and weaknesses.

C) Taking a risk in grey areas.

D) Using past experience for any action.

44. The benefit of team work is:
A) Co-operation B) Co -ordination C) Concentration D) All of the above

45. The main problem that people face in a team is:
A) Group Think B) Conflict C) Stability D) Cohesiveness

46. Team acquaints and establishes ground rules. In which stage of team formation formalities are preserved and members are treated as strangers?
A) Storming B) Performing C) Forming D) Adjourning

Read the following and answer Question numbers 57-60:

Arjun is an engineering student. He has one class that he finds difficult. The instructor has formed permanent class teams with weekly case studies to present to the class and a final team project to complete. He is the leader of that team. He dislikes relying on others for a final grade and gets frustrated trying to keep the team members focused on their tasks. Some people are late for meetings, others don't do their share of the work, and two team members have personality conflicts with each other.

47. What should Arjun do as the team leader?
A) Be Dominating.
B) Be Aggressive.

C) Motivate others to co operate.
D) Do all the work himself.

48. How should conflict between two team members be handled by them?
A) By Ignoring each other.
B) By Arguing.
C) By showing empathy.
D) By holding grudges.

49. How can team members contribute better to the task?
A) By being work shirkers
B) By having role clarity
C) By having role conflict
D) By depending on the leader

50. One disadvantage of teamwork shown in this situation is:
A) Non Cooperation
B) Disagreements
C) Politics
D) Stalemates

Read the following and answer Question numbers 51-53:

You went to a party with some people but the person who was driving had too much drink and refused to let anyone else drive. Other friends were hesitating to speak up fearing the driver's reaction where as one person started shouting at the driver and they were about to get into a fight.

51. Being Afraid to speak up in such a critical situation is a sign of being:

A) Assertive B) Aggressive C) Submissive D) Competitive

52. Shouting and fighting with someone in a critical situation like this one is a sign of being:

A) Assertive B) Aggressive C) Submissive D) Competitive

53. How you should handle this situation?

A) By keeping Quiet
B) By joining the fight
C) By ignoring that the driver is drunk
D) By clearly and confidently telling the driver that it's not safe for him to drive.

Choose the correct option, True or False:

54. Our interpretation of inferiority or superiority relative to others will influence our sense of capability and worth.

A) True B) False

55. A person can't be self-confident and diffident at the same time.

A) True B) False

56. The disadvantages are very limited compared to the large number of advantages we have with teams and teamwork.

A) True B) False

57. What is an effective group discussion?

A) One person talking and others listening carefully throughout the discussion.
B) The discussion is not dominated by any one person
C) Everybody trying to speak together to put their point forward.
D) Only few candidates talking and covering the topic.

58. During a group discussion be aware of people's reactions and feelings, and try to respond appropriately.

A) True B) False

59. While participating in a group discussion, the candidate should:

A) Keep the tone low and use simple words.
B) Use encouraging body language and tone of voice, as well as words.
C) Try not to make eye contact with the other participants.
D) Keep his mouth shut as much as possible

60. What skills are considered by the observer while the group discussion goes on?

A) Grooming and Interpersonal Skills
B) Leadership and Reasoning Skills
C) Assertiveness and Teambuilding Skills
D) All of the above.

61. Which of the following is true as a ground rule of a group discussion?

A) Do not speak until you are asked to do so.
B) You can be aggressive if others are not listening to your point.
C) Complain the observer if others do not let you speak.
D) Don't be defensive if someone disagrees with you.

62. Ramesh was participating in the group discussion for the first time. There were 12 participants and the topic given was quiet controversial. Ramesh was not very confident about the topic given. What was the best step for him to take?

A) Take the initiative to start the topic.
B) Be patient and listen to others to gather some information about the topic and then speak.
C) Keep quiet all throughout the group discussion.
D) Repeat the exact points given by others.

63. Which of the following phrases one should use to become a part of the group discussion?

A) I would like to add
B) I totally disagree with you

C) Please excuse me but this is funny
D) I want to tell you

64. Which of the following is not true for good presentation skills?

A) Have the slides ready and in order.
B) Test the visual to see if people at the back of the room can see it.
C) Stand to the side of the screen and face the audience.
D) Look at the screen all the time and not at the audience.

65. To make your presentation impressive, you should:

A) Fill all the slides with lot of content.
B) Keep the ratio of visuals more than content.
C) Keep the ratio of content more than visuals.
D) Try to fit in all facts and figures and content in the slides.

66. Which of the following can hamper your presentation?

A) Being brief and clear in giving the subject and purpose.
B) Being lively and enthusiastic.
C) Using a variety of media sources.
D) Reading from the slides.

67. If you are asked to summarize a GD, it means the GD has come to an end. Do not add anything more.
A) True B) False

68. A short story is a good way to initiate a group discussion.
A) True B) False

69. Which of the following is not important for presentation skills?

A) Good Voice
B) Content & Visual Aids
C) Audience Management
D) Audience Management

70. Dressing up for your office you socks should match the color of your ___.

A) Shirt B) Tie C) Trousers D) Shoes

71. It is considered good manners to answer the phone.....

A) As soon as it rings B) Anytime we feel convenient

C) After 4 rings D) Within 3 rings.

72. Rahul has dressed up in western formals for an important client meeting. He is wearing a brown belt, so he should wear _____ color shoes.

A) Blue B) Black C) Brown D) Grey

73. The optimum length for the tie is _____.

A) Below the belt.
B)Two inches above the belt.
C) Just touching the belt.
D) None of the above.

74. Rohan, who has to appear for his first interview has gone to purchase a set of formal clothes. He has bought a pair of black pants and now he should buy a _____ colour shirt to look appropriately dressed for the interview.

A) Black B) Navy blue C) Off-white D) Dark Green

75. Neha has got lovely long hair and she never ties them on any occasion. Since Neha has to appear for an interview, she should:

A) Leave her hair flowing free.
B) Make a neat bun.
C) Colour her hair with golden streaks.
D) Get it trimmed.

76. No matter what type or level of job you apply for, you should never greet the receptionists and secretaries with a smile as it is considered misbehavior.

A) True
B) False

77. While attending an evening office dinner with the top boss present-

A) Wear only formal dress
B) Wear lots of accessories
C) Wear only simple salwar kameez
D) Be daring in your dress and make up

78. When addressing an audience in a formal suit with a 3-button coat-

A) All three buttons of the coat should buttoned
B) Only the top two buttons should be buttoned
C) The second and third button should be buttoned
D) All the buttons should be unbuttoned

79. When attending any meeting or interview, cover yourself nicely with a loud perfume or a deodorant.

A) True
B) False

80. Before meetings and Interview, one should avoid foods which may leave breath odour. Use a breath mint if needed.

A) True
B) False

81. Which of the following is not one of the suggested strategies for preparing for your job interview?

A) Conduct research on the company/industry

B) Prepare answer to possible interview questions

C) Gather key resources (resumes, educational certificates, reference list)

D) Research salaries so you can ask about salary and benefits at the beginning of the interview.

82. Being the most qualified candidate for the position just about guarantees you will get the job.

A) True
B) False

83. Which part of the interview is the most important?

A) First few minutes
B) Answering the toughest questions
C) Final minutes
D) The whole time

84. The best thing to do in an interview when you get a question that stumps you temporarily is:

A) Sit there and just stare at the interviewer

B) Keep saying "good question, good question"

C) Respond with, "I just really can't answer that"

D) Paraphrase the question while giving yourself time to think.

Q. 85-86
Ankit is a final year student and much before he could start preparation for campus placement, the placement season has kick started. The first campus hiring is going to begin tomorrow. It'll decide Ankit's future -

85. What is the first thing that Ankit should take care of?
A) Resume
B) Academic Transcripts
C) Dress code
D) Salary

86. Ankit wonders as to what his resume should talk about?
I) Education-Qualification and Credentials
II) Involvement in Extra- curricular Activities
III) Hobbies
IV) Project Experience

Choose the correct option from A to D:

A) I, III, IV
B) II, III, IV
C) I, IV
D) I, II, III, IV

87. It's the summer time. What should Ankit do to avoid sweaty odour?
A) Use perfume
B) Use cologne
C) Use deodorant
D) Use cotton fabrics

88. Should Ankit wear bright colors such as red and green for the interview?
A) Yes
B) No

89. Ankit recently bought a pair of athletic shoes and considers them appropriate for the interview. Are athletic shoes appropriate for an interview?
A) Yes
B) No

90. What are three most important keys to success in an interview that Ankit needs to remember?

A) Good perfume/cologne, nice smile, fresh breath

B) Making eye contact, showing enthusiasm, speaking clearly

C) Fresh breath, nice smile, making eye contact

D) Developing rapport, fresh breath, good perfume/cologne

91. The Process of Communication ends with:

A) Receiver B) Message C) Sender D) Feedback

92. _____ refers to the interpretation of message at the receiver's end.

A) Encoding B) Paraphrasing C) Decoding D) Summarising

93. The external obstacles are distortions which exist outside both the sender and receiver, and prevent effective communication. These are called-

A) Filters B) Barriers C) Channels D) Means

94. Which of the following in not a barrier?

A) Distance B) Jargon C) Noise D) Pre-conceived notions

95. Which of the following in not a filter?

A) Language B) Anger C) Disappointment D) Shyness

96. Which of the following does not help in overcoming barriers and filters?

A) Paraphrasing B) Clarifying C) Interrupting D) Empathy

97. Percentage impact of tone of voice in communication is -

A) 28 B) 38 C) 48 D) 55

98. A message can only be deemed effective when it is-

A) Repeated back as proof of understanding.
B) Understood by others and produces the intended results.
C) Delivered with confidence.
D) Communicated face-to-face.

99. Which of the following statement is False?

A) Barriers and filters hinder effective communication.
B) Communication is a process.
C) A message sent by the sender can be interpreted only in one way.
D) Communication can be verbal or non-verbal.

100. In the communication process, to encode means to -

A) Speak to large groups of people.
B) Translate ideas into a code.
C) Block a pathway between the sender and receiver of a message.
D) Interpret a code.

Read the following and answer question number 101 to 103:

Surya is all excited about the latest film he has seen. He wants to discuss the film with his best friend. He tries to speak to his friend over the telephone. When his friend finds out that Surya wants to chit chat about the film, he tells him that since he has some urgent work, they could talk about it later when they meet in college. Surya's enthusiasm is lost and feels dejected at his friend's behavior.

101. What is the main reason of the negative outcome in this situation?

A) Physiological factors
B) Surya's friend's behaviour
C) Noise in communication
D) Surya's behaviour

102. What is the filter in this case?

A) Enthusiasm
B) Disinterest
C) Rigid Attitude
D) Superiority Complex

103. What Surya could have done to handle the situation better?

A) Be aggressive
B) Be persistent
C) Showed empathy
D) Showed sympathy

Directions for Q.Nos. 104-105: In each of the following questions, a part or the complete sentence has been underlined. You have to find the right way of writing the underlined part of the sentence.

104. If some Indians look at where they are going, it can be seen that our goal is money.

A) If some Indians look at where they are going, it can be seen that our goal is money.
B) If some Indians look back to where they are going, it can be seen that our goal is money.
C) If some Indians look at where they are going, it can be seen that their goal is money.
D) If some Indians look at where they are going, they can see that their goal is money.

105. The entire cast and crew of the film, enjoyed splashed in the pool , bathing in the ocean , and,particularly , to sun bathe on the shore.

A) The entire cast and crew of the film, enjoyed splashing in the pool, bathing in the ocean, and, particularly, to sun bathe on the shore.
B) The entire cast and crew of the film, enjoyed splashing in the pool, to have a bath in the ocean, and, to sun bathe on the shore.
C) The entire cast and crew of the film, enjoyed swimming in the pool to bathe in the ocean, and, particularly sun bathing on the shore.
D) The entire cast and crew of the film, enjoyed swimming in the pool, bathing in the ocean, and, particularly, sun bathing on the shore.

Directions for Q.Nos.106-110: Fill in the blanks with the correct option:

106. The meeting had to be _____ because of the strike.
A) Set off B) put off C) laid off D) set to

107. The boss wants an explanation. How do you _____ the decrease in sales?
A) Make up B) break down C) add up D) account for

108. Either my father or my brother _____going to sell the house.
A) is B) are

109. The news from the front _____ bad.
A) is b) are

110. Our department _____ get any bonus pay when we move.
A) doesn't B) don't

Printed Pages-3 **University Roll No.....**

B. Tech. II Year IV Semester, End Term Examination, 2013-14

CEE-201: Environmental Studies

Time: 02 ½ Hours **Total Marks: 40**

Section-A

Note: Attempt All Questions **1 X 16 = 16 Marks**

I. Who introduced the term ecosystem?

II. What are the four components of ecosystem?

III. Define the term balanced ecosystem?

IV. Define the term fluorosis.

V. What is desertification?

VI. What do you mean by abstract type natural resource?

VII. Define endemic species.

VIII. Define ex-situ and in-situ biodiversity conservation.

IX. How can you measure the noise?

X. What is population explosion?

XI. What do you mean by global warming?

XII. Write ambient air quality standards in respect of noise in silence zone.

XIII. What is full form of SHAPE?

XIV. Explain the term ecological footprint?

XV. When India enacted forest conservation act?

XVI. Which commission introduced the term sustainable development in the report "our common future"?

Section-B

Note: Attempt Any Four Questions **3 x 4 = 12 Marks**

I. Write major roles of government in the protection of environment.

II. Give objectives of environmental protection act.

III. Describe the guidelines for excellence in environmental education.

IV. Write notes on Bombay Natural History Society.

V. Briefly describe the LCA.

Section-C

Note: Attempt Any Three Questions **4 x 3 = 12 Marks**

I. Discuss the salient features of the wild life protection act 1972.

II. What is EIA? Discuss the steps involved in it.

III. Explain the scopes of the subject environmental ethics.

IV. What are the major obstacles in the path of sustainable development?

Printed Pages-4 **University Roll No.....**

B. Tech. II Year IV Semester, End Term Examination, 2013-14

ITE-221: Database Management Systems

Time: 02 ½ Hours **Total Marks: 40**

Section-A

Note: Attempt all questions. **1x16=16 Marks**

I. What do you understand by spurious tuple?

II. What do you understand by union compatibility?

III. What are the roles of DBA?

IV. What do you understand by participation constrains? Explain both.

V. What is difference between database systems vs. file systems?

VI. Differentiate between stored vs derived attribute

[For Q7 to Q9 only] Consider relation R(A,B,C,D), the FDs are $A \rightarrow B$, $A \rightarrow C$, $A \rightarrow D$, and $B \rightarrow A$.

VII. Find the candidate keys of R

VIII. List transitive dependencies in R (assume any CK as PK).

IX. Find the highest current normal form of R.

X. Define functional dependency.

XI. Difference between primary key and candidate key.

XII. Name five aggregate functions in SQL

XIII. Discuss atomicity in transaction.

XIV. Define exclusive lock and its use.

XV. Write the conditions for conflict instructions.

XVI. Discuss vertical fragmentation.

Section-B

Note: Attempt any four questions. **3x4=12 Marks**

I. Discuss view serializability.

II. What are issues of deadlock handling in databases? Discuss the centralized approach of deadlock handling.

III. Write the compatibility matrix of lock modes IS, IX, S, X.

IV. Write the conditions under validation based protocol.

V. What the major issues of distributed database.

Section-C

Note: Attempt any three questions. **4x3=12 Marks**

I. Consider the following interleaved execution of two transactions:

Time	Transaction T_1	Transaction T_2
t_1		Begin tr.
t_2	Begin tr.	sum=0
t_3	Read (bal(X))	Read(bal(X))
t_4	bal(X)=bal(X)-10	sum=sum +bal(X)
t_5	Write(bal(X))	Read(bal(Y))
t_6	Read(bal(Z))	sum=sum+bal(Y)
t_7	bal(Z)=bal(Z)+10	
t_8	Write(bal(Z))	
t_9	Commit	Read(bal(Z))
t_{10}		sum=sum+bal(Z)
t_{11}		Commit

Transaction T_1 is transferring amount 10 from account X to account Z, whereas T_2 is summing the balance in the three accounts

(i) Identify the concurrency problem associated with the above schedule by monitoring the balance amount in accounts X, Y, and Z and the sum at each time step. Explain why the problem occurred.

(ii) Use 2PL to resolve the problem.

Assume accounts X, Y, and Z had Rs. 100, 50, and 25 resp. before the start of transactions.

II. Consider the following sequence of instruction of a schedule:

$R_2(A)$; $R_1(B)$; $R_1(C)$; $R_5(E)$; $R_5(F)$; $R_6(G)$; $R_2(B)$; $W_2(B)$; $W_3(C)$; $R_1(D)$; $R_6(H)$; $R_4(B)$; $W_4(B)$;$R_4(C)$; $W_4(C)$; $R_1(D)$; $W_1(D)$;

[Hint: $R_2(A)$- means Transaction T2 reads data item A]

Draw the precedence graph and give the serializability order for above schedule.

III. Compare the strategies of distribution (Centralized, Fragmented, Full replication Selective replication) in terms of local reference, reliability and availability, storage, performance and communication cost) Use tabular format.

IV. What is R_timestamp and W_timestamp of data items. How it is useful in Time stamp ordering protocol(TSOP). State the conditions under TSOP to write data items in transactions.

Printed Pages-3 **University Roll No.....**

B. Tech. II Year IV Semester, End Term Examination, 2013-14

CSE 221: Operating Systems

Time: 02 ½ Hours **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16 Marks)**

I. Define Operating Systems.

II. Write the fundamental difference between asymmetric and symmetric multiprocessing.

III. List the four major categories of the benefits of multithreaded programming.

IV. Differentiate between I/O-bound and CPU-bound processes.

V. What is a bootstrap program, and where is it stored?

VI. What do you mean by degree of multiprogramming?

VII. What effect does the size of the time quantum have on the performance of an RR algorithm?

VIII. What role does the dispatcher play in CPU scheduling?

IX. Differentiate between preemptive and non preemptive scheduling.

X. What is the Critical Section Problem?

XI. What are the three general ways that a deadlock can be handled?

XII. What is the difference between deadlock prevention and deadlock avoidance?

XIII. What do you mean by virtual memory?

XIV. What is thrashing?

XV. Define Belady's Anomaly.

XVI. Define Latency and seek time with respect to disk I/O.

Section B

Note: Attempt any Four Questions **(3x4=12 Marks)**

I. Given a memory partitions of 100K, 500K, 200K, 300K, 600K (in order), how would each of the First-fit, Best-fit and Worst-fit algorithm place processes of 212K, 417K, 112K AND 426K (in order)? Which algorithm makes the efficient use of memory?

II. In a paged segmented system, a virtual address consists of 32 bits of which 12 bits are used for displacement, 11 bits are a segment number and 9 bits are a page number. Calculate. **a.** Page size **b.** Maximum number of pages **c.** Maximum segment size **d.** Maximum number of segments.

III. In a demand page memory 200 ns are required to satisfy memory request if the page is in memory. If a page is not in memory the request takes 7ms, then calculate the effective access time if the page fault rate is 5%.

IV. Differentiate between internal and external fragmentation. Which one occurs in paging? Which one occurs in segmentation?

V. Explain various types of directory structures with their merits and demerits.

Section C

Note: Attempt any Three Questions **(4x3=12 Marks)**

I. What is paging? Why is it used? What is the hardware support required for paging? Describe this with the help of a diagram.

II. How many page faults will occur for the following reference string with 4 page frames using LRU and Optimal Page Replacement Algorithms.

1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2

III. Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at cylinder 125, the queue of pending requests in FIFO order is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from current head position what is the total distance that the disk arm moves to satisfy all the pending requests for SSTF and LOOK disk scheduling algorithms.

IV. Differentiate between –

- Programmed I/O and Interrupt driven I/O.
- Absolute path name and Relative path name

Printed Pages-4 **University Roll No.....**

B. Tech. II YEAR IV Semester, End Term Examination, 2013-14

CSE223: Computer Graphics

Time: 02 ½ Hours **Total Marks: 40**

Section-A

Note: Attempt All Questions. **1 x 16=16 marks**

I. Consider two raster systems with resolution of 640 * 480 pixels and 1280 * 1024 pixels. How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second?

II. List out the merits and demerits of DVST?

III. A line connecting the points (1, 1) and (5, 3) is to be drawn, using the DDA algorithm. Find the values of x and y increments.

IV. The inputs to the boundary-fill algorithm are the co-ordinates of an interior point (x, y). So, which co-ordinate represents for boundary color and which co-ordinate represent for fill color?

V. Find the clipping co-ordinates for a line p_1p_2 where $p_1 = (10, 10)$ and $p_2 = (60, 30)$ against window with $(x_{wmin}, y_{wmin}) = (15, 15)$ and $(x_{wmax}, y_{wmax}) = (25, 25)$.

VI. Differentiate between convex and concave polygons by using examples.

VII. Scale the polygon with co-ordinates A (2, 5) and B (7, 10) by two units in x direction and two units in y direction.

VIII. Find a transformation of a triangle A (1, 0), B (0, 1), C (1, 1) by translating one unit in x and y direction and then rotating 45° about the origin.

IX. Give the 3D transformation matrix for:

a) Translation b) Rotation

X. Write a short note on perspective projection.

XI. List the advantages of B-splines over Bezier curves.

XII. What are the different ways of specifying spline curve?

XIII. Assuming the Depth-buffer algorithm allows 128 depth values levels to be used, how much memory would a 512 * 512 pixels display require to store the Depth buffer. If the scene consists of 14 objects, what is the frame buffer memory requirement?

XIV. What is Hypertext?

XV. List out the building blocks of multimedia.

XVI. Differentiate between lossy and lossless image compression.

Section-B

Note: Attempt Any Four Questions. **3 x 4=12 marks**

I. How to create hypermedia messages? Give an example. Also explain hypermedia message components.

II. List the advantages and disadvantages of Back-face detection, Depth-Buffer method and A Buffer method.

III. What are the issued analyses Multimedia Data Interface Standards?

IV. Write short notes on GKS/ PHIGS Graphics Standards.

V. Find the total number of bits required to represent a text (symbols with frequency are given in table) by using Huffman coding algorithm:

Symbols	Frequency
A	30
B	30
C	15
D	15
E	10

Section-C

Note: Attempt Any Three Questions. **4 x 3=12 marks**

I. Discuss in detail about Digital Video Interactive (DVI) with suitable diagrams.

II. Suppose a message composed of five symbols need to be communicated over a communication channel. The symbols and their probabilities of occurrence are shown in the table. By using Shannon Fano algorithm **compute the following**: SF-Code for characters, Average length of characters, Entropy of the source and Redundant bits/ symbol in SF code.

Symbols	Probability
a1	0.05
a2	0.15
a3	0.04
a4	0.50
a5	0.26

III. Explain in detail the Real Time Multimedia architectural framework.

IV. What is compression? Write all the steps for JPEG image compression process in detail.

Printed Pages-3 **University Roll No.....**

B. Tech. II YEAR IV Semester, End Term Examination, 2013-14

CSE222: Introduction to Microprocessors

Time: 02 ½ Hours **Total Marks: 40**

Section- A

Note: Attempt All Questions. **1 x 16 = 16 marks**

I. Write the use of Direction Flag.

II. The no. of address lines required to address a memory of size 32 K is.....

III. A Bus cycle is equal to how many clock periods.

IV. Which microprocessor pins are used to request and acknowledge a DMA transfer?

V. What do you mean by Pseudo instructions?

VI. Which type of JMP instruction assembles if the distance is 0020 H bytes?

VII. The processor of knowing the status of device and transferring the data with matching speeds is called.....

VIII. What is Op-code?

IX. What are the names of the four segment registers?

X. Explain DAA instruction.

XI. In which condition over flow flag is set in 8086.

XII. Name the internal registers of 8086.

XIII. Write the two inputs signals of 8254 programmable timer counter.

XIV. IRR stands for....

XV. Which interrupts has highest Priority.

XVI. Name all the hardware interrupts?

Section- B

Note: Attempt Any Four Questions. **3 x 4 = 12 marks**

I. Write and explain the three data transfer mechanism in computer system.

II. Explain the parallel I/O mode of 8255?

III. Write and explain different mode of operation of 8253?

IV. Discuss the following mode of operation of 8259

i) Cascade mode ii) Buffered mode iii) Nested mode

V. Specify the number of times the following loops are executed. Also explain the flow of execution.

i)
```
        MVI  A, 17H
LOOP:   ORA   A
        RAL
        JNC  LOOP
```

ii)
```
        MVI  A, 17H
LOOP:   RAL
        ORA  A
        JNC  LOOP
```

iii)
```
        LXI  B, 1000H
LOOP:   DCX B
        NOP
        JNZ  LOOP
```

Section- C

Note: Attempt Any Three Questions. **4 x 3 = 12 marks**

I. Explain the data transfer types supported by 8237 and also write the bit pattern of command register of 8237A?

II. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

III. Explain the data transfer types supported by 8237 and also write the bit pattern of command register of 8237A?

IV. Discuss the organization of USART with a functional block diagram.

Printed Pages-3 **University Roll No.....**

B. Tech. II Year IV Semester, End Term Examination, 2013-14

CSE 224: Computer Networks

Time: 02 ½ Hours **Total Marks: 40**

Section-A

Note: Attempt All Questions. **1X16=16marks**

I. What is the difference between intranet and extranet?

II. A binary signal is sent over a 3-khz channel whose signal-to-noise ratio is 20dB. Calculate the maximum achievable data rate?

III. What are the drawbacks of bus topology?

IV. What is the difference between a LAN and a WAN?

V. A noiseless 4-KHz channel is sampled every 1 msec. What is the maximum data rate?

VI. Before each Ethernet frame, there is a 64-bit preamble. What is the purpose of this?

VII. Why is a network switch preferred over a hub?

VIII. Convert the IP address 197.228.17.56 into binary?

IX. Calculate the maximum number of class A and B Network Ids.

X. How FDDI offers higher reliability than token ring protocol?

XI. Where is the special IP address 127.0.0.0 used?

XII. Suppose that an Ethernet LAN is used to serve a group of 10 stations. How much bandwidth is available to each station if the 10 stations are connected to a 100 Mbps Ethernet Hub?

XIII. Bits are being streamed onto a particular link. The link can accept a new bit every 20 nsec. What is the link's bandwidth?

XIV. Explain the difference between symmetric and asymmetric encryption.

XV. During a TCP communication session, if the packets arrive to the destination out of order, what will happen to the original message?

XVI. How can a NAT box improve security?

Section-B

Note: Attempt Any Four Questions. **(3x4=12marks)**

I. On which layer retransmission of lost data can be done. Discuss the pros and cons of doing it at each layer?

II. How does an email server works? Discuss.

III. What are the fundamental services provided by the transport layer? How does TCP implement each of these services?

IV. Explain Three-Way Handshake Mechanism used by TCP to terminate a Session reliably.

V. Define flow control. How does flow control at transport layer differ from flow control at data link layer?

Section-C

Note: Attempt any three questions. **(4x3=12marks)**

I. Briefly describe the SNMP (Simple Network Management Protocol) architecture. Your description should cover the following: manager, query, client and traps.

II. TCP uses buffers on both the sending and the receiving side. Describe the purpose of buffer at each side.

III. What is domain name? How is a domain name translated into equivalent IP address? Explain with the help of an example.

IV. Discuss MIME protocol used for e-mailing.